MIL-V-48773 (AR)
AMENDMENT 2
15 March 1985
SUPERSEDING
AMENDMENT 1
25 September 1978

MILITARY SPECIFICATION

VANADIUM PENTOXIDE

(FOR USE IN AMMUNITION)

This amendment forms a part of Military Specification MIL-V-48773, dated 19 June 1975, and is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Delete the preamble and substitute the following:

"This specification is approved for use by the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense."

PAGE 2

3.1.3 Chemical Properties. Delete table in its entirety and substitute the following table:

FSC:6810

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"Table I: Chemical Properties

Property	Requirement	Applicable Paragraph
Moisture Content, %, max	0.40	4.4.3.3
Vanadium (as V_2O_5, % min (on Moisture Free Basis)	99.50	4.4.3.4
Halogens (as Cl, Br, I), %, max	0.007	4.4.3.5
Iron, (as Fe_2O_3), %, max.	0.015	4.4.3.6
Sodium (as Na_2O), %, max.	0.025	4.4.3.7
Resistivity - ohm-cm @ 4000 lbs.	7,500	4.4.3.10"

3.1.4 Particle size. Delete in its entirety.

PAGE 4

4.4.2.1 Sampling. Add the following to the end of the second sentence:

"or the samples from each batch that is combined into a shipping lot shall be subjected to the tests listed below in accordance with MIL-STD-105, Level II."

Delete the table and substitute the following table:

"Table II - Classification of Defects

Inspection/Requirement	Defect Classification
Form (see 3.1.1)	Major
X-ray Diffraction (see 3.1.2)	Major
Properties (see 3.1.3)	Major
Moisture	
Vanadium (as V_2O_5)	
Halogens (as Cl, Br, I)	
Iron (as Fe_2O_3)	
Sodium (as Na_2O)	
Resistivity	

4.4.3.1 Form. In the first sentence delete " 1/2 pound" and substitute "100 g."

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4.4.3.5 Chlorides. Delete this paragraph in its entirety and substitute the following:

"4.4.3.5 Halogens (as Cl, Br, I) - Determine the halogens present in the compound by use of a standard analytical technique for halogens which has prior approval (see 6.5)".

4.4.3.7 Sodium. To end of paragraph add the following sentence: "The method used shall have prior approval (see 6.5").

4.4.3.8 Silica. Delete this paragraph in its entirety.

4.4.3.9 Particle Size. Delete this paragraph in its entirety.

4.4.3.10 Add a new paragraph and number it 4.4.3.10.

"4.4.3.10 Resistivity

4.4.3.10.1 Equipment.

a. Laboratory Press - force indicated for 1500, 2000, 3000 and 4000 pounds.

b. Height fixture.

c. Test Cylinder & Pistons

d. Keithley 602 Electrometer

e. 5K Ohm and 10K Ohm Resistors

f. Laboratory Balance

Assemble height fixture and set up on laboratory press. Assemble Test Cylinder & Pistons and place between plates of height fixture. Connect Electrometer leads and close press until zero ohms is indicated (pistons are in contact). DIAL INDICATOR SHOULD READ 0.500. ADJUST FOR THIS READING.

Periodically check electrometer by measuring the 5K ohm and 10K ohm resistors.

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4.4.3.10.2 Procedure. The cylinder and bottom piston shall be assembled. One gram of V_2O_5 shall be poured into the cylinder (Be sure there is an even layer of V_2O_5 in the cylinder). The top piston shall be inserted into the cylinder and placed into the press. The electrometer leads shall be connected to the top and bottom pistons. A pressure of 1500 pounds shall be applied. Record resistance (ohms) and dial indicator reading. A pressure of 2000, 3000 and 4000 pounds shall be repeated and recorded. The pressure shall be released. Remove the cylinder, pistons and V_2O_5 pellet. Repeat the above as necessary.

4.4.3.10.3 Calculations.

a. The piston diameter ("0.500") calculates to 1.2668cm^2 area.

b. Subtract Dial Indicator readings from 0.500. Convert the results to centimeters by multiplying by 2.54.

c. Use the following formula:

$$\text{Resistivity} = \frac{AR}{L} = \frac{1.2668R}{L}$$

where:

A = Area of piston of test fixture
R = Resistance (ohms)
L = Thickness of sample (cm)
and Resistivity = (ohm cm)."

* 6.3 Add the following source to the end of the first sentence:

"and Metallurg Inc., 25 East 39th St., New York, NY 10016."

6.5 Add the following new paragraph:

"6.5 The method used to determine the requirement should be submitted to AMSMC-QAR-R through the contracting officer for prior approval".

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, delections) from the previous

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amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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Preparing Activity
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Project (6810-AB44)

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